One Million Users, One Service

Why large services use many machines

SCAN / LEARN

THE MAIN IDEA

A popular service is usually a coordinated fleet, not one magical server. Traffic distribution prevents overload, but capacity is finite.

FOLLOW THE SYSTEM

01 USERS

Many people send requests at once.

02 FRONT DOOR

A load balancer routes requests.

03 SERVER FLEET

Several healthy servers process requests.

04 FEEDBACK

Health checks and scaling react to demand.

MANY REQUESTS → LOAD BALANCER → SERVERS

COMMON MISCONCEPTION

Adding servers makes any service infinitely fast.

MORE ACCURATE

Databases, networks, budgets and shared state still create limits.

THREE QUESTIONS TO KEEP

PROBLEM: How do we serve many people at once?

FAILURE: A bottleneck creates slow replies or errors.

IN REAL LIFE: Streaming, shopping, public portals.

SOLO DISCOVERY

Try alone: draw three servers; route nine paper requests to the least-busy server each time.